Technology Stack

Phase 2 | Selected tools and their roles

Stack inventory

Layer	Technology	Role
Frontend	React 18, Vite 5	Client-rendered application and development/build tooling.
Routing	React Router	Home, account, builder, saved trips, and trip-detail routes.
UI	Framer Motion, Lucide React, CSS	Motion and icons; project styles define visual presentation.
Backend	Python, FastAPI, Pydantic	HTTP API, dependency injection, request validation, typed response models.
Database	Supabase Postgres	Persistent users, trips, itinerary days, preferences, saved places, and history schema.
AI	Groq API or Ollama	Optional structured itinerary generation.
Fallback	Local curated/procedural Python services	Generates structured itinerary content without remote model access.
Auth	HS256 JWT and password hashing	Signed bearer access tokens and password verification.
Hosting	Docker Compose, Render, Vercel	Containerized setup and cloud deployment configuration.

Why the combination fits

- React and Vite support an interactive form-driven web interface.
- FastAPI and Pydantic provide a clear JSON API boundary and schemas.
- Supabase supplies managed Postgres when configured.
- A provider abstraction lets the backend choose remote generation while preserving a local fallback.
- Environment-based configuration supports local and hosted deployments.

Configuration and limitations

Backend dependencies are listed in backend/requirements.txt and frontend packages in frontend/package.json. Hugging Face-related configuration may be present, but a Hugging Face generation adapter is not implemented in the inspected AI service. The frontend uses an API base URL from VITE_API_BASE_URL when configured.

Prepared from the current project source and configuration. Product assumptions and proposed work are labeled; no user research, test result, or deployment result is implied unless explicitly stated.